# C-STGB: Real-Time Financial Surveillance Architecture & Conformal Triage Pipeline

Zero-Copy Streaming Ingestion → Continuous-Time Tri-Band GNN → Conformal CRC Triage → 3-Tier Operational Dispatch

## Streaming Ingest & Invariant

### Wire Protocols & Memory

SWIFT MT103 • ISO-20022 • Kafka

UTXO Blockchains (Bitcoin, Eth)

**DuckDB Micro-Batches**

**Apache Arrow Zero-Copy Buffer**

Causal Ordering: $t_e \leq t_{\text{eval}}$ (No Leak)

### 12-D Physical Invariants ($\mathbf{z}_{\text{inv}}$)

Mass Flow: $\Phi_{\text{flow}} = \ln(1 + \frac{A_{\text{out}}}{A_{\text{in}} + \epsilon})$

Hawkes: $\lambda_u(t) = \mu_u + \sum \alpha e^{-\rho \Delta t}$

Directed Taint: $s_{\text{fwd}}, s_{\text{bwd}}, [\mu_A, \sigma_A^2]$

### Dynamic Subgraph Sampler

Top-$K$ Causal Neighborhood ($K \leq 15$)

**Tensor Output:** $\mathbf{x}_u \in \mathbb{R}^{d_0}$, $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$

Fast-Path 1-Hop Cache Hit: 99.4%

**Ingestion Latency SLA: 0.11 ms**

## C-STGB Spatio-Temporal Neural C

### 3-Band Temporal GNN & Edge Gate

Tri-Band Attention: $\tau_1 = 2.4\,\text{h}, \tau_2 = 24\,\text{h}, \tau_3 = 168\,\text{h}$

Learnable Edge Gate: $\hat{g}_{ij} = \sigma(\mathbf{w}_g^\top [\mathbf{h}_i \,\|\, \mathbf{h}_j \,\|\, \mathbf{e}_{ij}])$

**Prunes 65.9% Chaff Camouflage** ($\hat{g}_{ij} \geq \delta_{\text{floor}}$)

### Typology SMOTE & Adaptive Fusion

Typology Latent SMOTE: $\mathbf{h}_{\text{syn}} = (1-\rho)\mathbf{h}_u + \rho \mathbf{h}_v, \ (u, v \in \mathcal{C}_k)$

Evidence Fusion: $\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1-\alpha_u)[\mathbf{z}_{\text{inv}} \,\|\, \mathbf{x}_u]$

**Cold-Start Guarded: 89.60% F1 for** $\deg(u) \leq 2$

### Cost-Sensitive Bayes Risk Head

Asymmetric Risk: $\tau^* = \arg\min_{\tau} (15 \cdot \text{FN} + 1 \cdot \text{FP})$

**Calibrated Decision Threshold:** $\tau^* = 0.18$ **(15 : 1 Cost Ratio)**

Posterior Risk Output: $\hat{p}_u = \sigma(\mathbf{w}^\top \mathbf{h}_u^* + b) \in [0, 1]$

**Inference Latency SLA: 0.16 ms**

## Conformal Triage

**Class-Cond. CRC Gate**

$1 - \alpha \geq 99\%$

$\hat{q}^{(0)}, \hat{q}^{(1)}$

$\mathcal{D}_{\text{cal}}$ Holdout

## Tier 1: High-Risk Quarantine

Fraud $\{1\}$

**Immediate Quarantine Hold • Wire Freeze**

Automated FinCEN Form 111 SAR Dossier Draft

GNNExplainer Subgraph • SHA-256 Merkle Receipt

**Operational Rate: 18 / 10,000 tx (0.18%)**

## Tier 2: Compliance Review Desk

Review $\{0\}$

**4-Eyes AML Review Desk • Senior Examiner Queue**

Abstention Set: Ambiguous Boundary Triage

Active Learning Queue • Workload Cut: -99.51%

**Operational Rate: 49 / 10,000 tx (0.49%)**

## Tier 3: Straight-Through Clear

Clear $\{0\}$

**Automated Wire-Speed Release (STP Clearing)**

Zero Manual Friction • Continuous Risk Guard

Bounded Selective Risk: Residual FN < 0.13%

**Operational Rate: 9,933 / 10,000 tx (99.33%)**

## REAL-TIME PRODUCTION SLAS • STATISTICAL CONVERGENCE GUARANTEES • REGULATORY GOVERNANCE

**Latency Performance**
**0.45 ms Fast-Path**
P99: 0.82 ms • Throughput: 31,200 tx/s
Strict compliance with <10 ms SLA

**Workload Reduction**
**-99.51% Queue Cut**
Only 49 of 10,000 tx need review
85.7%–95.8% fewer false alerts

**Statistical Safety**
**≥ 99.0% Coverage**
Simultaneous minority class safety
0.00% empty sets • Avg size: 1.005

**Governance & Audit**
**OCC & SR 26-2 Ready**
Zero-hallucination graph evidence
Cryptographic SHA-256 Merkle trail